

CanvasPic: An Interactive Tool for Freely Generating Facial Images Based on Spatial Layout

Jiafu Wei
weijf21@mails.jlu.edu.cn
Jilin University
Chang Chun, China

Xi Yang*
yangxi21@jlu.edu.cn
Jilin University
Chang Chun, China

Chia-Ming Chang
info@chiamingchang.com
The University of Tokyo
Tokyo, Japan

Takeo Igarashi
takeo@acm.org
The University of Tokyo
Tokyo, Japan

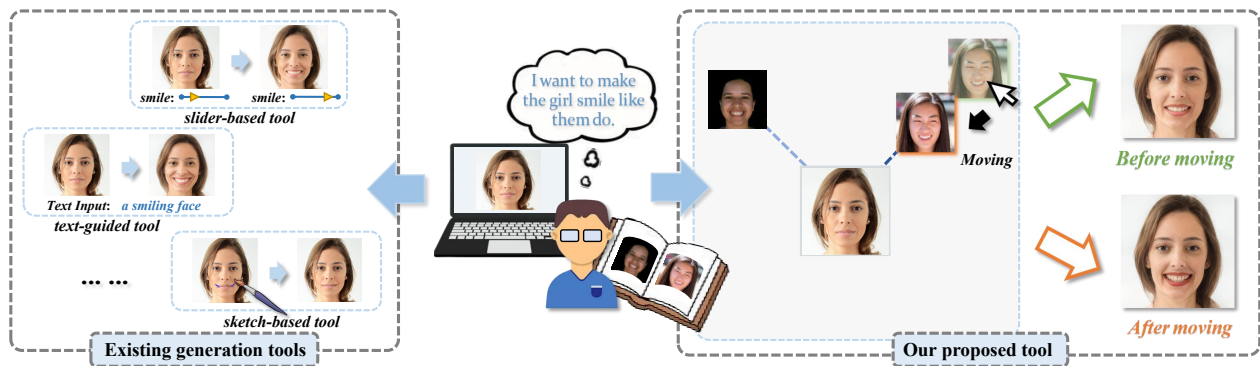


Figure 1: We introduce CanvasPic, a novel generation tool with a 2D spatial layout that allows users to import real-world images as references and generate desired results by adjusting the distances and attributes of these reference images.

ABSTRACT

In real-world usage, existing GAN image generation tools come up short due to their lack of intuitive interfaces and limited flexibility. To overcome these limitations, we developed CanvasPic, an innovative tool for flexible GAN image generation. Our tool introduces a novel 2D layout design that allows users to intuitively control image attributes based on real-world images. By interacting with the distances between images in the spatial layout, users are able to conveniently control the influence of each attribute on the target image and explore a wide range of generated results. Considering practical application scenarios, a user study involving 24 participants was conducted to compare our tool with existing tools in GAN image generation. The results of the study demonstrate that our tool significantly enhances the user experience, enabling more effective achievement of desired generative results.

*Corresponding author. Engineering Research Center of Knowledge-Driven Human-Machine Intelligence, MoE, Jilin Province Key Laboratory of Ancient Chinese Script, Culture relics and Artificial Intelligence.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3650952>

CCS CONCEPTS

• **Human-centered computing** → *Systems and tools for interaction design; Interactive systems and tools.*

KEYWORDS

CanvasPic, Human-centered AI, Interactive System, User Interface, GAN, Interview

ACM Reference Format:

Jiafu Wei, Chia-Ming Chang, Xi Yang, and Takeo Igarashi. 2024. CanvasPic: An Interactive Tool for Freely Generating Facial Images Based on Spatial Layout. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3613905.3650952>

1 INTRODUCTION

Despite the impressive capabilities of Generative Adversarial Networks (GANs), controlling the style and content of their generated images remains a challenge [8, 26]. Providing active control over GAN-generated image style and content is vital for meeting specific requirements in practical applications. This has led to the emergence of GAN image generation tools that empower users to express their creativity and ideas. Furthermore, personalized generation has gained recognition, resulting in various tools aiming to fulfill users' aspirations.

However, the existing tools still lack control flexibility and user-friendliness. For example, sketch-based [14, 27] tools often demand users to possess specific drawing skills or image editing experience. Additionally, the generated images often do not achieve the desired level of realism. Slider-based tools [11, 34] offer limited choices, neglecting diverse user needs and creative expressions. Text-based tools [22, 23, 25] rely on abstract input, resulting in a gap between the user's expectations and the obtained results. Furthermore, some tools [6] suffer from intuitive efficiency issues, such as cluttered interfaces and complex operations. As users increasingly seek greater freedom and personalization in generative tools, traditional modes of tools have become inadequate to meet the demands of specific scenarios.

To address the aforementioned challenges, we introduce CanvasPic, a novel tool designed for flexible image generation. The primary goal of CanvasPic is to offer a multifunctional image generation platform that goes beyond traditional control functionalities. Through an innovative 2D layout design, users can effortlessly adjust image positions to control generated results, enhancing both intuitive interaction and efficiency. This method grants users more control over results, elevating their creative freedom. Additionally, CanvasPic integrates real-world images as references, enabling users to guide the generation of target images using attributes from existing images. This not only aids result anticipation but also inspires creative exploration and experimentation, fostering a stronger user-content connection. We generate facial images in this paper which is one of the most general domains, however, our tool is not limited to facial images.

The main contributions of our work are summarized as follows:

- (1) We developed CanvasPic, a GAN-based tool that empowers users to leverage attributes from real-world images for controlling the generation results of portrait images.
- (2) We introduce a novel 2D spatial layout interface that enables users to control image attributes through distance adjustments. This design fosters user exploration and anticipation of generated results, enhancing task intuitiveness.
- (3) We conducted a user study involving 24 participants to evaluate user satisfaction with CanvasPic. The results of the user study demonstrate the significant impact of CanvasPic in enhancing the overall user experience.

2 RELATED WORK

2.1 Advancing Image Generation: From GANs to Real-World Synthesis

In the field of image generation, prominent techniques include the Diffusion Model [30], VAE [19], and GAN [10]. Each exhibits distinct strengths: Diffusion Model excels in producing high-quality, realistic natural images; VAE is suitable for image reconstruction and variation generation, while GAN performs well in generating synthetic images closely resembling real ones, particularly in attribute control [39, 41]. Leveraging the characteristics of GAN, various image generation technologies based on GAN have been developed, including the influential StyleGAN [15]. For example, researchers [9, 13] proposed new generation methods using latent code transformation. Khodadadeh et al. [17] achieved successful generation tasks by mapping latent vectors to target images.

Liu et al.'s STGAN [24] improved attribute manipulation accuracy through selective transfer, while He et al.'s AttGAN [12] introduced attribute classification constraints for high-quality generation. However, these methods have difficulty controlling real-world images.

In recent years, research has shifted toward applying image-generation techniques to real-world scenarios. This led to the development of GAN inversion [28, 32], including text-guided [22, 23] and sketch-based methods [14, 27, 40]. Especially, Patashnik et al.'s StyleCLIP [25] merged StyleGAN and CLIP models, enabling text-driven image generation based on users' textual descriptions. Ko et al. [21] devised a GAN-based communication system for efficient interactions and sketch adjustments among webcomic authors and artists.

Moreover, there has been a proliferation of text-based generative inpainting methods in recent years [33, 42, 44]. These approaches utilize textual descriptions to guide the image inpainting process, amalgamating text descriptions with image generation to fully exploit the strengths of both modalities. In contrast to traditional methods solely reliant on textual guidance, integrating local image information enables a more precise comprehension of user intent and requirements, thereby facilitating semantically and contextually enhanced image inpainting.

However, existing image generation tools still suffer from limitations in control flexibility and user-friendliness, making them difficult to effectively utilize in practical scenarios.

2.2 Enhancing User Experience Through Spatial Layouts

The concept of spatial layout has broad applications and is often used to define relationships among icons, objects, and information [5, 29, 38, 43]. Bauer et al. [2] summarized the concept of a "pile" to showcase the benefits of structured spatial layouts. Yi et al. [31] introduced the "Dust & Magnet" technique, utilizing magnet metaphor and interactive methods to enhance the understanding and accessibility of multidimensional data. Watanabe et al. [35] applied spatial layout to design an interface for object space aggregation to streamline information management.

Well-designed spatial layouts significantly enhance task efficiency and user experience [36, 37]. For example, Kleiman et al. [20] introduced a system enabling users to interact with a large image collection on a 2D canvas based on nearest neighbors in a high-dimensional feature space, resulting in efficient task interaction. Brown et al. [3] proposed a system that allows experts to directly interact with visual data representation, ensuring scalability in data dimensions and streamlined task processing. Additionally, Chang et al. [4] designed a spatial layout annotation interface to improve image annotation quality by non-expert annotators.

3 TOOL: CANVASPIC

Our CanvasPic is an image generation tool based on 2D spatial layout (see Figure 2). Users import target and reference images, specify attributes, and manipulate the reference images in the workspace to control the generated result. We have formulated the following design goals for our tool: **Goal 1:** Allow users greater flexibility in generating diverse results. **Goal 2:** Provide users with a simplified

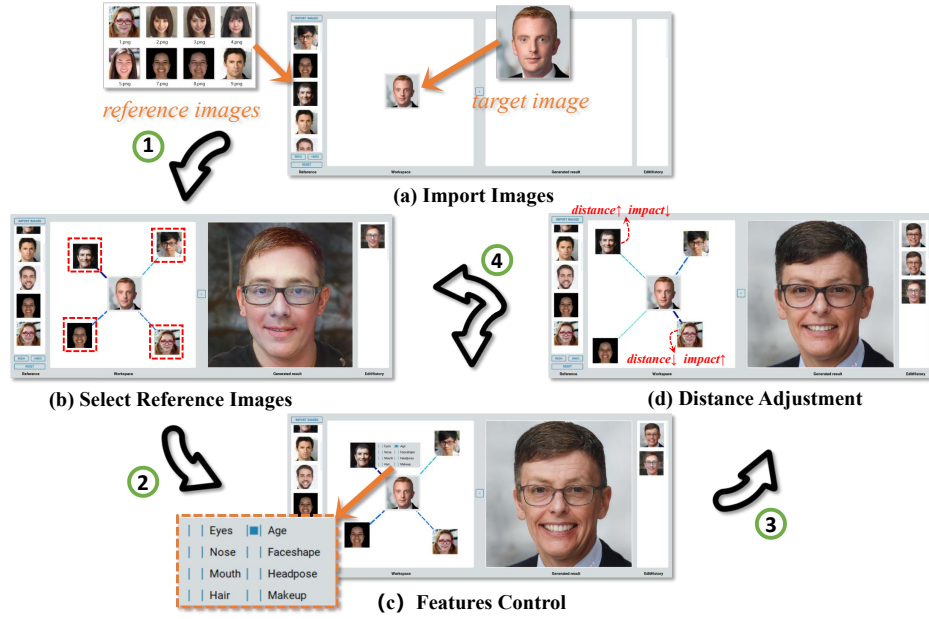


Figure 2: CanvasPic workflow: (a) Importing reference images and the target image. (b) Placing reference images within the spatial layout workspace. (c) Selecting attributes from the reference images. (d) Controlling the generated results by adjusting the distance between the images. Users can then return to step (b) or (c) to continue the image generation process.

and intuitive generative experience through a user-friendly interface layout. **Goal 3:** Offer users easily accessible and efficient image control methods. Additional details are available in the supplementary material.

3.1 Interface Design and Functions

Our CanvasPic comprises four components: Reference image bar, Spatial layout workspace, Results panel, and History module (see Figure 3).

Reference image bar. Users can import images from the real world or provided datasets as reference images through the reference image bar (**Goals 1 & 3**). The reference image is used to provide attributes for the generated image.

Spatial layout workspace. For CanvasPic, we introduce a novel 2D spatial layout (**Goal 2**). Users import their desired image (target image) by clicking the central area of the spatial layout workspace. Then, users can add reference images by clicking or dragging them from the reference image bar to the workspace, and connection lines are also added to show their relationships. Right-clicking on a reference image triggers an attribute selection box to allow users to select their desired attributes from the image (**Goal 3**). In our implementation, the attribute selection box is designed to contain eight attributes that are the most critical for facial generation. They are categorized into two groups: local attributes (including eyes, nose, mouth, and hair) and global attributes (including age, face-shape, headpose, and makeup). These attributes in the selection box can be expanded for different tasks.

After that, users can move the reference image to control the influence of selected attributes on the generated results (**Goal 3**).

The closer the reference image is to the target image, the more similar they are, and vice versa. We allow users to redo, undo, or reset their adjustments during the generation process by clicking the appropriate buttons. Additionally, we correspondingly change the color and thickness of the connecting line to further visualize the impact (**Goal 2**).

Results panel. After users finish their actions in the workspace, they can click the right arrow button to generate a high-resolution image, which will be displayed in the result panel.

History module. We also add a history module that stores all the generated results (**Goal 2**). Users can click on the generated image to return to the history point.

3.2 Algorithm

We utilize pre-trained e4e [32] and StyleGAN2 [16] as encoder and generator respectively.

Attribute transfer. Due to the distinct characteristics of local and global attributes, we have explored two algorithms for transferring them individually. For local attributes, we employ a mask preprocessing method inspired by techniques found in StyleMapGAN and r-FACE [7, 18], supplying corresponding masks for each reference image. However, the aforementioned algorithm is not applicable to global attributes. Therefore, we resort to an image code subtraction method inspired by the approach presented in Image2StyleGAN [1].

Distance weight calculation. To realize adjusting the distance between the reference image and the target image impacts the generated result, our tool calculates this distance automatically and assigns an inverse attribute weight to the reference image.

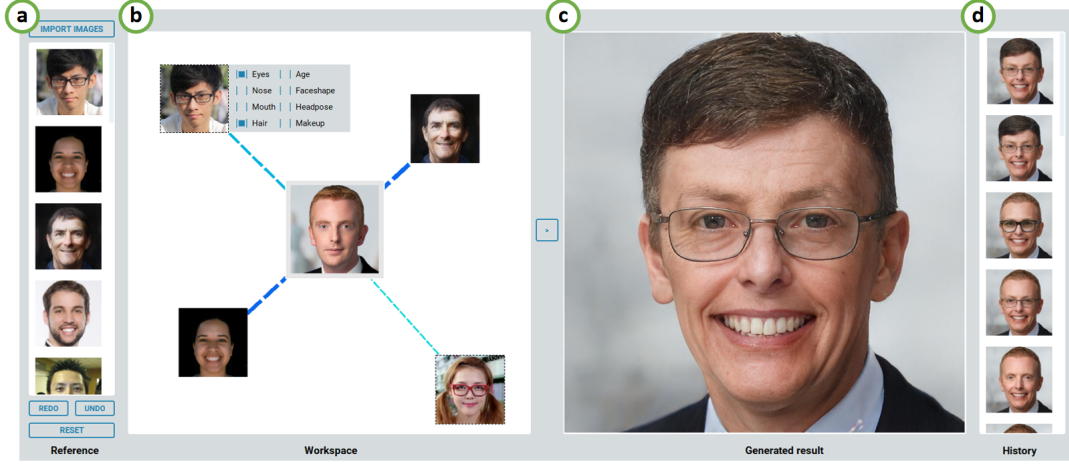


Figure 3: Our interface. The interface comprises four components: a) Reference image bar, b) Spatial layout workspace, c) Results panel, and d) History module.

Additional algorithmic details can be found in the supplementary material.

4 EXPERIMENTS

Existing generation tools fall into three primary categories: slider-based, text-guided, and sketch-based tools. We chose representative tools from each category: HFGI [34], StyleCLIP [25], and SketchEdit [40]. These three tools exemplify the core and widespread forms found within generative tools. HFGI adopts the control mechanism of the commonly used one-dimensional slider tool. StyleCLIP, on the other hand, is among the pioneering tools to employ text-based control for image generation, attracting considerable attention. Meanwhile, SketchEdit represents a foundational form within sketch-based tools. Our experiments focused on local and global attribute image generation tasks, utilizing images from the FFHQ dataset [15].

4.1 Local Attribute Generation

We used four tools to add smiles to portraits (see Figure 5). HFGI, a slider-based tool, reveals limitations associated with such tools for specific tasks. One-dimensional sliders offer restricted control options that limit user control capabilities. Consequently, HFGI can only adjust smiles to a singular style.

SketchEdit, despite offering considerable control freedom, presents some distinct issues. Firstly, the generated images exhibit poor quality, with blurry regions and lacking proper light and shadow effects. Secondly, the tool is challenging to operate, particularly for users without drawing skills. Lastly, there is a significant gap between the drawing effects and generation results, leading to notable discrepancies between the generated images and users' expectations.

In contrast, the remaining tools provide more flexible generation options by incorporating external reference information, liberating users from predefined constraints. Users can broaden the scope of control by utilizing diverse reference images or input sentences. For example, users can obtain distinct smile effects by introducing

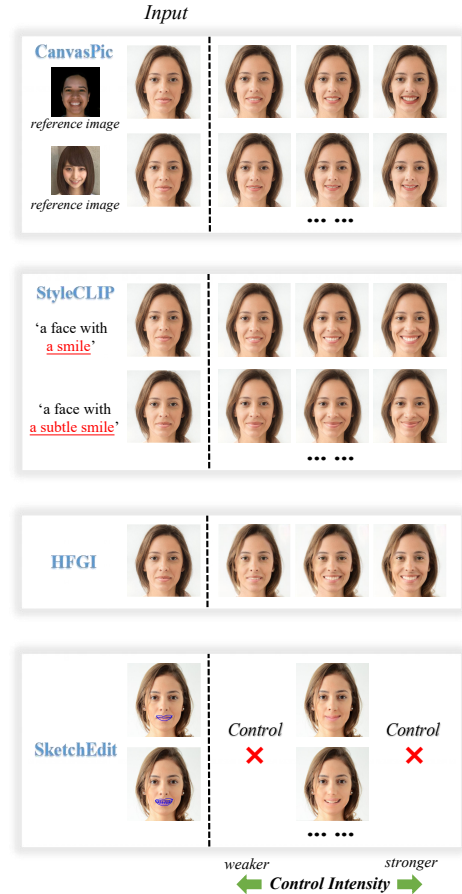


Figure 4: Comparison of the results generated by different influence intensities of local attributes. CanvasPic utilizes image distance for intensity adjustment, while StyleCLIP and HFGI use sliders, SketchEdit cannot adjust intensity.

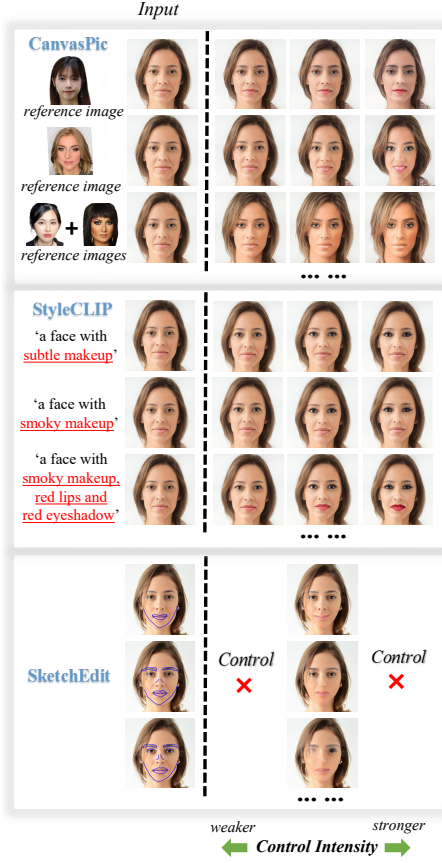


Figure 5: Comparison of the results generated by different influence intensities of global attributes.

reference images with desired expressions or by describing their intended effects through input sentences.

4.2 Global Attribute Generation

We conducted experiments on global attribute generation, specifically focusing on applying makeup effects to portraits. Three tools were employed for this task: our proposed tool, StyleCLIP, and SketchEdit (see Figure 5). HFGI was excluded due to its inability to support makeup generation. Similar limitations were observed in both StyleCLIP and SketchEdit, particularly when dealing with abstract attributes such as makeup effects, resulting in unsatisfactory results.

Furthermore, while using StyleCLIP for image generation, we observed remarkably similar generation results regardless of the input text. This hampers users from achieving the desired results and constrains their creativity. Additionally, obtaining specific generation results in StyleCLIP often necessitates detailed textual descriptions, which introduces challenges such as potential oversight of details, as illustrated by the red eyeshadow in Figure 5. Currently, accurately conveying intent through text remains a task that is relatively challenging for users.

In contrast, we believe our proposed tool provides a more intuitive and user-friendly way of image generation compared to existing tools. Users can easily transform images into desired styles by referencing real-world images. Our tool eliminates the need for complex descriptions and offers users a straightforward means to convey their intentions.

5 USER STUDY

We conducted a user study to assess the strengths of our proposed tool and its impact on user experience. Following the tests, we conducted semi-structured interviews and distributed questionnaires to gather user feedback.

5.1 Study Design

Tool Selection. We selected four existing tools: CanvasPic (Ours), HFGI, StyleCLIP, and SketchEdit. To ensure objectivity and anonymity, we assigned distinct code names to each tool. CanvasPic was denoted as “Reference Generate” (RG), HFGI as “Slider Generate” (SG), StyleCLIP as “Text Generate” (TG), and SketchEdit as “Sketch Edit” (SE).

Participants. We recruited 24 participants, half male and half female, aged between 20 and 30. All participants have experience using editing software such as Photoshop or Meitu Xiu Xiu, mainly for portrait editing. Experiment durations were set at a maximum of 100 minutes for male participants and 120 minutes for females. After the experiment, each user receives a payment of \$5.

Tasks. We established three generation themes: designing suitable glasses for the portrait (*Theme A*), adding appropriate makeup to the portrait (*Theme B*), and altering the portrait to your desired look (*Theme C*).

Please note that due to the general lack of experience among males with makeup, therefore male participants are not involved in *Themes B*, while female participants participate in all three themes. Additionally, HFGI and SketchEdit are exclusively designated for *Themes C*. Consequently, female participants use CanvasPic (*Themes A, B, and C*), StyleCLIP (*Themes A, B, and C*), HFGI (*Themes C*), and SketchEdit (*Themes C*) for their tasks, male participants use CanvasPic (*Themes A and C*), StyleCLIP (*Themes A and C*), HFGI (*Themes C*), and SketchEdit (*Themes C*) for their tasks. Each generation task has a maximum duration of 6 minutes.

Preparation. We provided participants with pre-selected images (target images & reference images) and presented them with three designated image generation themes.

Target images. The original images we provide to participants. They use images corresponding to their gender, with males using one specific image and females using another specific image. This differentiation aims to promote participants’ sense of identity and minimize gender distinctions, while also facilitating a more diverse range of experimental results.

Reference images. There are a total of 28 reference images, which are only used in CanvasPic, and all participants utilize the same set of reference images.

Interview. We conducted semi-structured interviews to capture participants’ experiences and feedback (see Tabel 1). We employed the following evaluation criteria: (Q1) *Degree of flexibility*; (Q2) *Ease of implementation*; (Q3) *Effortlessness*; (Q4) *Intuitiveness*; (Q5)

Table 1: Quantitative results on our questions. The results indicate overall higher satisfaction with CanvasPic among participants.

Comparison tool	Q1		Q2		Q3		Q4		Q5		Q6		Q7	
	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD
<i>CanvasPic (Ours)</i>	4.50	0.71	4.29	0.89	3.83	1.11	4.21	0.91	4.29	0.74	3.96	0.98	4.42	0.64
<i>StyleCLIP</i>	3.21	1.00	3.38	1.07	3.08	1.38	3.33	1.21	3.58	0.91	4.04	0.74	3.50	0.96
<i>HFGI</i>	2.58	1.08	2.83	0.99	3.54	1.04	3.17	1.43	2.83	0.99	3.46	1.26	2.79	1.19
<i>SketchEdit</i>	2.54	1.29	1.88	1.09	2.54	1.50	2.63	1.25	2.17	1.21	1.92	1.08	2.25	1.23

Generation effect; (Q6) Harmony; (Q7) Willingness to use. The Likert five-point scale was employed by the participants to respond to each criterion.

5.2 Quantitative Results

In this section, we use the abbreviation P to designate participants. For example, we represent the first participant labeled as P1. All statistical tests are conducted at a significance level (α) of 0.05.

We used the Friedman test to analyze our data. For Q1, twenty-one participants regarded CanvasPic as having the highest degree of flexibility ($M=4.50$, $SD=0.71$) ($s = 36.38$, $p < .05$) (Table 1: Q1). Similarly, nineteen participants found CanvasPic to be highly accessible for implementation ($M=4.29$, $SD=0.89$) ($s = 43.90$, $p < .05$) (Table 1: Q2). Regarding Q3, sixteen participants perceived the tool to exhibit strong effortlessness ($M=3.83$, $SD=1.11$) ($s = 12.89$, $p < .05$) (Table 1: Q3). Among the twenty-four participants, eighteen reported that CanvasPic demonstrated excellent intuitiveness compared to the other three tools ($M=4.21$, $SD=0.91$) ($s = 19.18$, $p < .05$) (Table 1: Q4). Concerning Q5, twenty participants expressed satisfaction with CanvasPic’s generative performance in comparison to the other three tools ($M=4.29$, $SD=0.74$) ($s = 39.75$, $p < .05$) (Table 1: Q5). Sixteen participants believed that CanvasPic’s generated results possessed the highest level of harmony ($M=3.96$, $SD=0.98$) ($s = 39.02$, $p < .05$) (Table 1: Q6), slightly inferior to StyleCLIP, which eighteen participants also found impressive ($M=4.04$, $SD=0.74$) (Table 1: Q6). Furthermore, twenty participants exhibited a stronger intention to use CanvasPic ($M=4.42$, $SD=0.64$) ($s = 37.61$, $p < .05$) (Table 1: Q7). In summary, CanvasPic yielded satisfactory results in terms of flexibility, ease of implementation, intuitiveness, and generative performance for the participants.

5.3 Qualitative Results

After each task, we conducted semi-structured interviews to learn more about the participants’ experiences.

Participants showed a clear preference for tools with high degrees of freedom and flexible control options. In the interviews, three-quarters of participants expressed a preference for tools with a high degree of freedom, and twenty-one believed that our tool offered the greatest level of freedom. The reasons include:

Facilitate goal achievement. P6 mentioned, “The RG tool offers me more control options, helping me achieve results that better align with the theme.” *Nurture creativity.* P4 stated, “High-degree-of-freedom tools showcase more possibilities, making me more interested in generating images and trying various combinations.”

Improve user experience. Specifically, six participants noted that high-degree-of-freedom tools provided smoother and more seamless control, significantly enhancing their usage experience.

Participants showed a clear preference for intuitive modes of image generation. The majority of participants preferred intuitive image generation tools, with eighteen finding our tool to be the most intuitive. The reasons include:

Generating images from real-world references is intuitively effective. P14 noted, “When dealing with thematic tasks, I have a certain idea of the generated image. I can use reference images to find matching attributes to achieve the desired result.” However, existing methods often use abstract controls such as preset sliders or text input, leading to significant discrepancies in results and user expectations. *The two-dimensional layout enhances intuitiveness.* P11 mentioned, “When performing multi-attribute control, the two-dimensional layout of the RG tool effectively indicates the attributes represented by reference images, allowing me to predict the generated results to some extent.”

6 DISCUSSION AND FUTURE WORK

6.1 Human-AI Collaborative Image Creation

AI-driven image generation tools hold immense potential in fostering collaboration between humans and artificial intelligence, creating new opportunities for synergy by merging human intuition and creativity with AI’s computational prowess. For instance, P17 mentioned, “While engaging in the task of designing glasses, I found myself creating styles I had never seen before. It’s interesting, and I believe it could hold significant value in the design field.”

Precisely and freely controlling image generation. In the realm of modern creativity and design, AI-driven interfaces have exerted a significant impact. Interactive tools enable users to more manageably and intuitively control the GAN-based image generation process. For instance, P2 stated, “The RG tool has excellent intuitiveness and control, making it easier for me to achieve desired results.” This aligns with the experimental results, indicating that CanvasPic provides users with greater creative freedom and expressive capacity, opening up broader possibilities for creators.

Impact of diverse input on user experience. Existing input methods can be broadly categorized into text and sketch inputs. In this paper, text input allows a certain degree of image control but exhibits limitations in fine adjustments and intuitiveness, sketching may require a certain level of artistic skill. Our proposed method, generating images using attributes from reference images, addresses limitations in both text and sketch methods. It avoids the ambiguity of text input and the skill requirements of sketching. Additionally, our method simplifies the input process, making it easier for users

to achieve their goals. Experimental results confirm that image input outperforms text and sketch input.

6.2 Future Work

Our current tool is specifically designed for close-up facial shots. In future updates, we aim to improve its versatility by training it on more diverse datasets, enabling it to handle various scenarios, including different parts of the human body, animals, and natural landscapes. We also plan to explore the strengths of different models in specific domains and integrate them to enhance the overall functionality of our tool. Currently, our method involves using masks to control local attributes, and the accuracy of these masks influences the quality of the generated results. In our upcoming work, we intend to incorporate lightweight object detection algorithms to automate and improve the accuracy of our method.

7 CONCLUSION

We introduce a novel GAN-based image generation tool, CanvasPic, aimed at providing users with enhanced freedom and intuitiveness in their creative experience. By incorporating a 2D spatial layout design, users can interact with elements within the layout, effortlessly adjust image attributes, and delve into exploring diverse potential results. Our user study indicates that CanvasPic outperforms existing tools in freedom, intuitiveness, user experience, and generated results. We hope that the results of this study provide insights into the future development of image generation tools.

ACKNOWLEDGMENTS

This work was supported by Jilin University (Grant No.419021422B08), JST ACT-X Grant Number JP-MJAX21AG and JST CREST Grant Number JP-MJCR17A1, Japan.

REFERENCES

- [1] Rameen Abdal, Yipeng Qin, and Peter Wonka. 2019. Image2stylegan: How to embed images into the stylegan latent space?. In *Proceedings of the IEEE/CVF international conference on computer vision*. 4432–4441.
- [2] Daniel Bauer, Pierre Fastrez, and Jim Hollan. 2005. Spatial tools for managing personal information collections. In *Proceedings of the 38th Annual Hawaii International Conference on System Sciences*. IEEE, 104b–104b.
- [3] Eli T Brown, Jingjing Liu, Carla E Brodley, and Remco Chang. 2012. Dis-function: Learning distance functions interactively. In *2012 IEEE conference on visual analytics science and technology (VAST)*. IEEE, 83–92.
- [4] Chia-Ming Chang, Chia-Hsien Lee, and Takeo Igarashi. 2021. Spatial labeling: leveraging spatial layout for improving label quality in non-expert image annotation. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–12.
- [5] Nan-Chen Chen, Jina Suh, Johan Verwey, Gonzalo Ramos, Steven Drucker, and Patrice Simard. 2018. AnchorViz: Facilitating classifier error discovery through interactive semantic data exploration. In *23rd International Conference on Intelligent User Interfaces*. 269–280.
- [6] Hai Dang, Lukas Mecke, and Daniel Buschek. 2022. GANSlider: How Users Control Generative Models for Images using Multiple Sliders with and without Feedforward Information. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–15.
- [7] Qi Yao Deng, Jie Cao, Yunfan Liu, Zhenhua Chai, Qi Li, and Zhenan Sun. 2020. Reference-guided face component editing. *arXiv preprint arXiv:2006.02051* (2020).
- [8] Omar Elharrouss, Noor Almaadeed, Somaya Al-Maadeed, and Younes Akbari. 2020. Image inpainting: A review. *Neural Processing Letters* 51 (2020), 2007–2028.
- [9] Lore Goetschalckx, Alex Andonian, Aude Oliva, and Phillip Isola. 2019. Ganalyze: Toward visual definitions of cognitive image properties. In *Proceedings of the IEEE/CVF international conference on computer vision*. 5744–5753.
- [10] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. 2014. Generative adversarial nets. *Advances in neural information processing systems* 27 (2014).
- [11] Erik Härkönen, Aaron Hertzmann, Jaakko Lehtinen, and Sylvain Paris. 2020. Ganspace: Discovering interpretable gan controls. *Advances in neural information processing systems* 33 (2020), 9841–9850.
- [12] Zhenliang He, Wangmeng Zuo, Meina Kan, Shiguang Shan, and Xilin Chen. 2019. Attgan: Facial attribute editing by only changing what you want. *IEEE transactions on image processing* 28, 11 (2019), 5464–5478.
- [13] Ali Jahanian, Lucy Chai, and Phillip Isola. 2019. On the “steerability” of generative adversarial networks. *arXiv preprint arXiv:1907.07171* (2019).
- [14] Youngjoo Jo and Jongyoul Park. 2019. Sc-fegan: Face editing generative adversarial network with user’s sketch and color. In *Proceedings of the IEEE/CVF international conference on computer vision*. 1745–1753.
- [15] Tero Karras, Samuli Laine, and Timo Aila. 2019. A style-based generator architecture for generative adversarial networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 4401–4410.
- [16] Tero Karras, Samuli Laine, Miika Aittala, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. 2020. Analyzing and improving the image quality of stylegan. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 8110–8119.
- [17] Siavash Khodadadeh, Shabnam Ghader, Saeid Motiian, Wei-An Lin, Ladislau Bölöni, and Ratheesh Kalarot. 2022. Latent to latent: A learned mapper for identity preserving editing of multiple face attributes in stylegan-generated images. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 3184–3192.
- [18] Hyunsu Kim, Yunjey Choi, Junho Kim, Sungjoo Yoo, and Youngjung Uh. 2021. Exploiting spatial dimensions of latent in gan for real-time image editing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 852–861.
- [19] Diederik P Kingma and Max Welling. 2013. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114* (2013).
- [20] Yanir Kleiman, Joel Lanir, Dov Danon, Yasmin Felberbaum, and Daniel Cohen-Or. 2015. Dynamicmaps: similarity-based browsing through a massive set of images. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. 995–1004.
- [21] Hyung-Kwon Ko, Subin An, Gwanmo Park, Seung Kwon Kim, Daesik Kim, Bohyoung Kim, Jaemin Jo, and Jinwook Seo. 2022. We-toon: A Communication Support System between Writers and Artists in Collaborative Webtoon Sketch Revision. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*. 1–14.
- [22] Jing Yu Koh, Jason Baldridge, Honglak Lee, and Yinfei Yang. 2021. Text-to-image generation grounded by fine-grained user attention. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*. 237–246.
- [23] Wenbo Li, Pengchuan Zhang, Lei Zhang, Qiuyuan Huang, Xiaodong He, Siwei Lyu, and Jianfeng Gao. 2019. Object-driven text-to-image synthesis via adversarial training. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 12174–12182.
- [24] Ming Liu, Yukang Ding, Min Xia, Xiao Liu, Errui Ding, Wangmeng Zuo, and Shilei Wen. 2019. Stgan: A unified selective transfer network for arbitrary image attribute editing. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 3673–3682.
- [25] Or Patashnik, Zongze Wu, Eli Shechtman, Daniel Cohen-Or, and Dani Lischinski. 2021. Styleclip: Text-driven manipulation of stylegan imagery. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2085–2094.
- [26] Fei Peng, Guo-lou Ping, and Shuang-kui Ge. 2019. A face privacy protection scheme using cnn based roi editing. In *2019 IEEE 21st International Conference on High Performance Computing and Communications; IEEE 17th International Conference on Smart City; IEEE 5th International Conference on Data Science and Systems (HPCC/SmartCity/DSS)*. IEEE, 345–352.
- [27] Tiziano Portenier, Qiyang Hu, Attila Szabo, Siavash Arjomand Bigdeli, Paolo Favaro, and Matthias Zwicker. 2018. Faceshop: Deep sketch-based face image editing. *arXiv preprint arXiv:1804.08972* (2018).
- [28] Elad Richardson, Yuval Alaluf, Or Patashnik, Yotam Nitzan, Yaniv Azar, Stav Shapiro, and Daniel Cohen-Or. 2021. Encoding in style: a stylegan encoder for image-to-image translation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2287–2296.
- [29] Frank M Shipman III, Catherine C Marshall, and Thomas P Moran. 1995. Finding and using implicit structure in human-organized spatial layouts of information. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 346–353.
- [30] Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. 2015. Deep unsupervised learning using nonequilibrium thermodynamics. In *International conference on machine learning*. PMLR, 2256–2265.
- [31] Ji Soo Yi, Rachel Melton, John Stasko, and Julie A Jacko. 2005. Dust & magnet: multivariate information visualization using a magnet metaphor. *Information visualization* 4, 4 (2005), 239–256.
- [32] Omer Tov, Yuval Alaluf, Yotam Nitzan, Or Patashnik, and Daniel Cohen-Or. 2021. Designing an encoder for stylegan image manipulation. *ACM Transactions on Graphics (TOG)* 40, 4 (2021), 1–14.

- [33] Su Wang, Chitwan Saharia, Ceslee Montgomery, Jordi Pont-Tuset, Shai Noy, Stefano Pellegrini, Yasumasa Onoe, Sarah Laszlo, David J Fleet, Radu Soricut, et al. 2023. Imagen editor and editbench: Advancing and evaluating text-guided image inpainting. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 18359–18369.
- [34] Tengfei Wang, Yong Zhang, Yanbo Fan, Jue Wang, and Qifeng Chen. 2022. High-fidelity gan inversion for image attribute editing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 11379–11388.
- [35] Nayuko Watanabe, Motoi Washida, and Takeo Igarashi. 2007. Bubble clusters: an interface for manipulating spatial aggregation of graphical objects. In *Proceedings of the 20th annual ACM symposium on User interface software and technology*. 173–182.
- [36] Jiafu Wei, Ding Xia, Haoran Xie, Chia-Ming Chang, Chuntao Li, and Xi Yang. 2024. SpaceEditing: A Latent Space Editing Interface for Integrating Human Knowledge into Deep Neural Networks. In *Proceedings of the 29th International Conference on Intelligent User Interfaces*.
- [37] Jiafu Wei, Haoran Xie, Chia-Ming Chang, and Xi Yang. 2022. Fine-tuning Deep Neural Networks by Interactively Refining the 2D Latent Space of Ambiguous Images. In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence Demo Track*. 5948–5951.
- [38] Jason Wu, Xiaoyi Zhang, Jeff Nichols, and Jeffrey P Bigham. 2021. Screen parsing: Towards reverse engineering of UI models from screenshots. In *The 34th Annual ACM Symposium on User Interface Software and Technology*. 470–483.
- [39] Ling Yang, Zhilong Zhang, Yang Song, Shenda Hong, Runsheng Xu, Yue Zhao, Wentao Zhang, Bin Cui, and Ming-Hsuan Yang. 2023. Diffusion models: A comprehensive survey of methods and applications. *Comput. Surveys* 56, 4 (2023), 1–39.
- [40] Yu Zeng, Zhe Lin, and Vishal M Patel. 2022. Sketchedit: Mask-free local image manipulation with partial sketches. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 5951–5961.
- [41] Fangneng Zhan, Yingchen Yu, Rongliang Wu, Jiahui Zhang, Shijian Lu, Lingjie Liu, Adam Kortylewski, Christian Theobalt, and Eric Xing. 2023. Multimodal image synthesis and editing: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2023).
- [42] Lisai Zhang, Qingcai Chen, Baotian Hu, and Shuoran Jiang. 2020. Text-guided neural image inpainting. In *Proceedings of the 28th ACM international conference on multimedia*. 1302–1310.
- [43] Weidong Zhang, Wei Zhang, and Jason Gu. 2019. Edge-semantic learning strategy for layout estimation in indoor environment. *IEEE transactions on cybernetics* 50, 6 (2019), 2730–2739.
- [44] Zijian Zhang, Zhou Zhao, Zhu Zhang, Baoxing Huai, and Jing Yuan. 2020. Text-guided image inpainting. In *Proceedings of the 28th ACM International Conference on Multimedia*. 4079–4087.